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DIQ project DATASET 12

PROJECT OF:
DATA AND INFORMATION QUALITY

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1 | Introduction

In the Big Data era, virtually every human interaction is transformed into data, a concept referred as '**datafication**' in the book Big Data Integration by Xin Luna Dong, Divesh Srivastava 2015 [5]. Concurrently, declining storage costs have made it economically feasible to retain this vast volumes of this data.

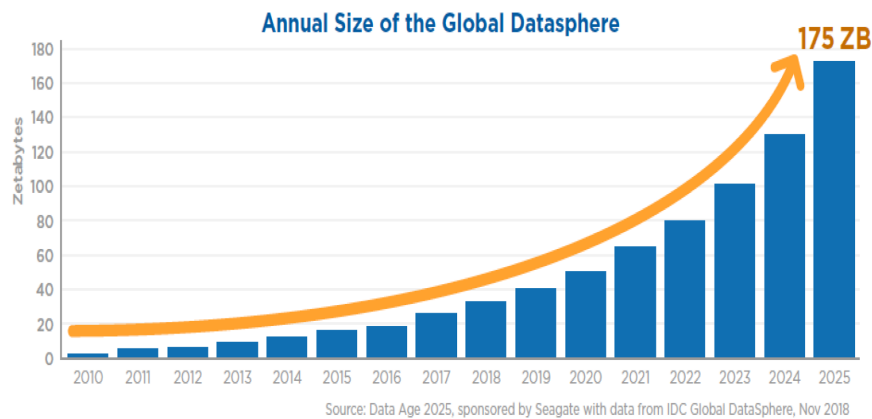


Figure 1.1: Data volume through years

Unfortunately, this sharp increase in data quantity has often come **at the price of data quality**, creating an immediate challenge for organizations seeking to leverage their data. This challenge of maintaining data integrity is further compounded by the recent advent of generative AI. While powerful, these models can contribute to a proliferation of low-quality or synthetic content, a phenomenon increasingly described as '**AI slop**'. Furthermore, attempts to use only AI to automatically fix data quality issues have often proven counterproductive [4], in many cases amplifying existing problems rather than solving them. As a result, the combination of massive data volume and new, unreliable automated processes creates an increasingly difficult environment for data-driven initiatives. As reported in various article we passed from 25% to upwards estimate of 40% of revenue affected by data quality issues in recent years.

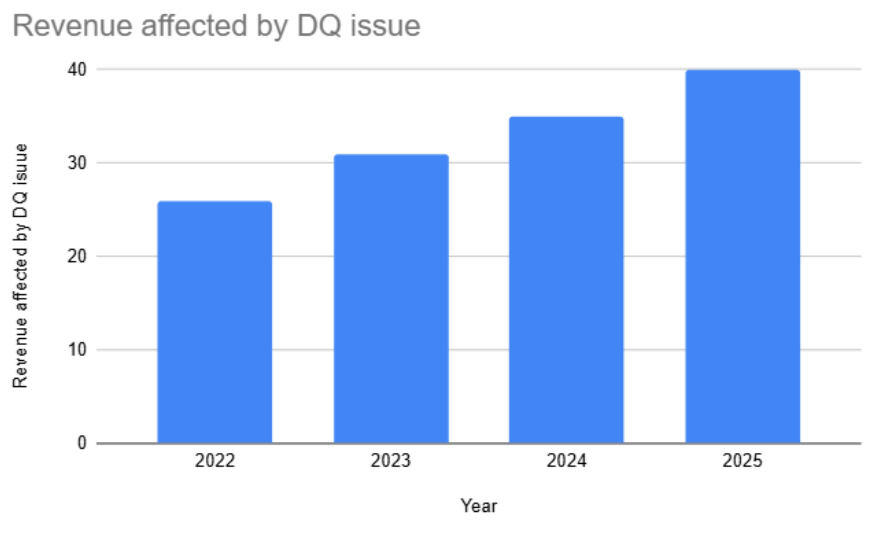


Figure 1.2: Revenue affected by DQ issue per year

This growing crisis in data quality is particularly concerning because the success of any data analysis or machine learning task fundamentally depends on the quality of the initial data. This core principle is succinctly encapsulated by the long-standing computer science adage, '**Garbage In, Garbage Out**' (GIGO).



Figure 1.3: GIGO

Recognizing this critical dependency, robust data governance plays an increasingly vital role in corporate strategy, as more companies acknowledge the pivotal importance of high-quality data in their predictive processes.

Given this context, it is essential to develop and refine effective methods for data cleaning. This project aims to address this very challenge directly. We will begin with a dataset exhibiting evident quality problems (which will be assessed in detail in subsequent sections) and apply a systematic cleaning process. The main goals are

- To transform this data from a potential liability into a valuable asset, ready for reliable analysis.

- To show in the process of doing so an effective data pipeline with possible strategic solutions to data quality issues we will encounter.

1.1. The dataset

The dataset "Comune di Milano Servizi alla persona parrucchieri estetisti" provides a comprehensive set of information regarding commercial activities in the beauty and personal care sector.

The dataset contains the following columns:

- **Tipo esercizio pa:** contains the type of activity based on the Italian ATECO code, e.g. "parrucchiere/estetista...". Data type=object
- **Ubicazione:** contains the full address of the activity location. Data type=object
- **Tipo via:** contains the type of street where the activity is located. Data type=object
- **Via:** contains the name of the street where the activity is located. Data type=object
- **Civico:** contains the street number of the activity location. Data type=object
- **Codice via:** contains an identifying code for the activity location. Data type=float 64
- **ZD:** a unique code assigned in Milan for different areas. Data type=object
- **Prevalente:** indicates the main activity performed at the location. Data type=object
- **Superficie altri usi:** the space dedicated to the activity area for personal staff only. Data type=float 64
- **Superficie lavorativa:** the effective space dedicated to the area open to the public. Data type=float 64

Figure (1.4) provides an overview of the dataset containing information on beauty and personal care activities in Milan, as described in the previous section.

df.head()

	Tipo esercizio pa	Ubicazione	Tipo via	Via	Civico	Codice via	ZD	Prevalente	Superficie altri usi	Superficie lavorativa
0	NaN	LGO DEI GELSOMINI N. 10 (z.d. 6)	LGO	DEI GELSOMINI	10	5394.0	6	NaN	NaN	55.0
1	NaN	PZA FIDIA N. 3 (z.d. 9)	PZA	FIDIA	3	1144.0	9	CENTRO MASSAGGI RILASSANTI NON ESTETICI	2.0	28.0
2	NaN	VIA ADIGE N. 10 (z.d. 5)	VIA	ADIGE	10	4216.0	5	CENTRO BENESSERE	2.0	27.0
3	NaN	VIA BARACCHINI FLAVIO N. 9 (z.d. 1)	VIA	BARACCHINI FLAVIO	9	356.0	1	TRUCCO SEMIPERMANENTE	NaN	NaN
4	NaN	VIA BERGAMO N. 12 (z.d. 4)	VIA	BERGAMO	12	3189.0	4	NaN	NaN	50.0

Figure 1.4: Overview of the dataset "Comune di Milano Servizi alla persona parrucchieri estetisti".

The dataset contains **3,909 rows**.

1.2. Libraries used

This project was developed in Python (v. 3.x) using Google Colab as the interactive development environment (IDE). The data preparation pipeline relied on several key libraries:

- **Pandas:** Utilized for high-performance data manipulation, cleaning, and analysis, primarily through its DataFrame structures.
- **NumPy:** Employed for efficient numerical computing and operations on multi-dimensional arrays.
- **YData Profiling:** Used to conduct automated Exploratory Data Analysis (EDA) and generate comprehensive data quality and statistical reports.
- **JSON:** The standard Python library used for parsing and handling data files stored in JSON format.
- **Matplotlib and seaborn:** used for plotting

2 | Implementation of the pipeline

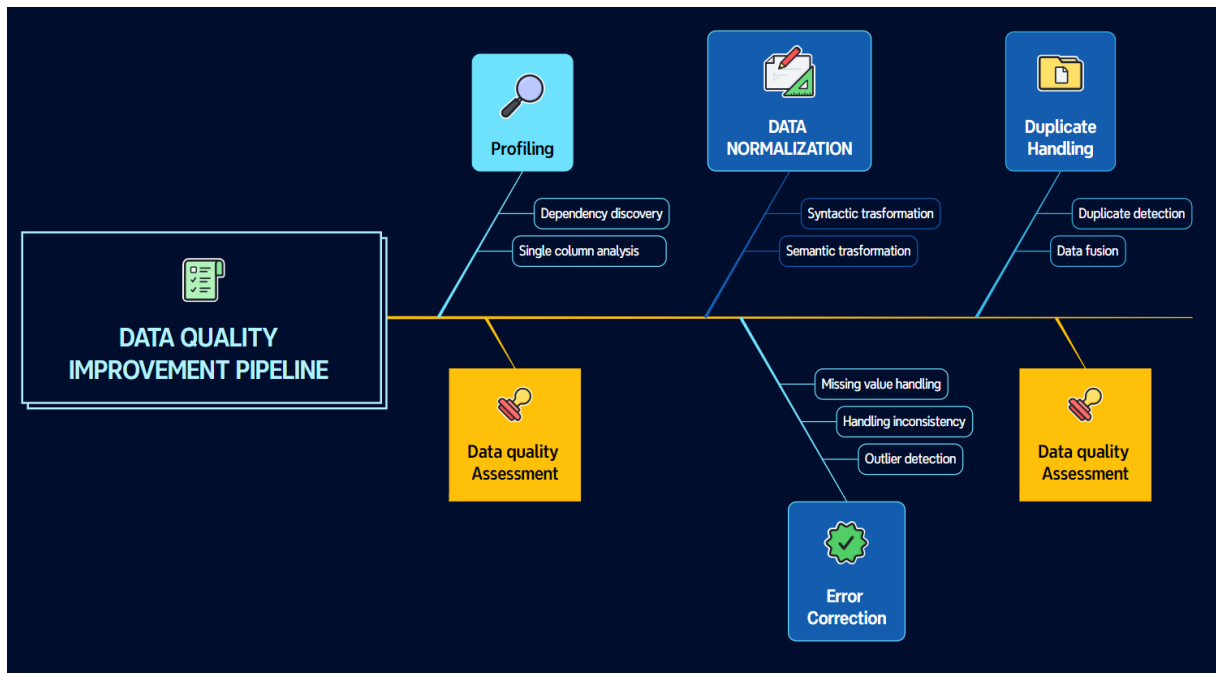


Figure 2.1: Pipeline

Data Profiling Data profiling consists in the systematic examination of the dataset's structure, format, and peculiar characteristics. This phase provides a comprehensive understanding of the data and allows the identification of potential anomalies or inconsistencies. Furthermore, it enables the formulation of an initial assessment of data quality and clarifies which aspects of the dataset require correction or transformation.

Data Cleaning The data cleaning phase aims to identify and eliminate inconsistencies, discrepancies, and errors within the dataset. This process builds upon the metadata and insights obtained during the data profiling phase. The main activities carried out during this stage include:

- **Standardization and Normalization:** Transform the data so that all semantically equivalent values are represented consistently. This may involve separating columns that refer to distinct concepts and merging categorical values that, although expressed differently, correspond to the same category.
- **Error Correction:** Address common data quality issues such as:

- Missing values
- Outliers
- Inconsistencies identified during the data profiling phase
- **Duplicate Detection:** Performed as a final step, since records referring to the same entity may initially appear different due to typos or inconsistent formatting. Once the standardization and correction processes are completed, duplicates can be more accurately identified and removed.

*Quality Assessment and Evaluation Upon completion of the cleaning process, and in accordance with the principles of the **Total Data Quality Management (TDQM)** cycle, the key data quality indicators previously defined are re-evaluated. This final assessment allows the measurement of tangible improvements in data quality and provides an overall evaluation of the effectiveness of the data preparation workflow.

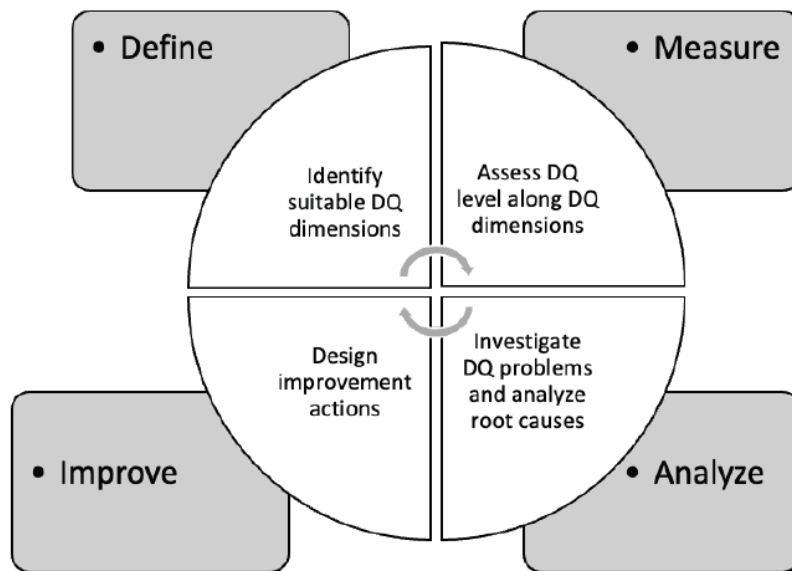


Figure 2.2: TDQM

2.1. Data profiling

2.1.1. Single column analysis

Let N_{total} be the total number of records in the dataset.

- **Actual:** This is the count of the number of records with an actual value (i.e., non-NULL and non-missing) for a given column.
- **Cardinality:** This is the count of the number of distinct actual values present in a column.

Derived Metrics

- **Completeness:** It measures the proportion of records that have a non-missing value.

$$\text{Completeness} = \frac{\text{Actual}}{N_{\text{total}}}$$

- **Uniqueness:** It measures the proportion of distinct values relative to the total number of records.

$$\text{Uniqueness} = \frac{\text{Cardinality}}{N_{\text{total}}}$$

- **Distinctness:** It measures the proportion of distinct values relative to the number of non-missing values. This is an indicator of value repetition among the existing (non-missing) values.

$$\text{Distinctness} = \frac{\text{Cardinality}}{\text{Actual}}$$

- **Constancy:** This metric is defined as the frequency of the most common value divided by the total number of values (N_{total}). It measures the degree to which a column's values tend to be the same.

$$\text{Constancy} = \frac{\text{Frequency of the Most Frequent Value}}{N_{\text{total}}}$$

Metric	Tipo esercizio pa	Ubicazione	Tipo via	Via	Civico	Codice via	ZD	Prevalente	Sup. altri usi	Sup. lavorativa
Missing	31	0	1	1	77	1	1	3615	3164	1308
Actual	3878	3909	3908	3908	3832	3908	3908	294	745	2601
Cardinality	103	3554	17	1370	235	1376	10	63	52	145
Completeness (%)	99.21	100.00	99.97	99.97	98.03	99.97	99.97	7.52	19.06	66.54
Uniqueness (%)	2.63	90.92	0.43	35.05	6.01	35.20	0.26	1.61	1.33	3.71
Distinctness (%)	2.66	90.92	0.44	35.06	6.13	35.21	0.26	21.43	6.98	5.57
Constancy (%)	27.02	0.10	76.20	1.07	6.50	1.07	15.43	26.53	13.83	4.61

Table 2.1: Data quality metrics for the dirty dataset

This initial analysis on all the columns is important to have an initial idea of the kind of data we are dealing with. From this we can start to notice for example the very low completeness value for the "Prevalente" field and the high constancy of "Tipo Via".

A more detailed analysis has been done with the support of the `ydata_profiling` library which generated a general profile of the dataset. This highlighted several interesting metrics for ensuring data quality for each attribute. We used the Y-data profiling and the metrics evaluated above as a starting point for a single column analysis that considered the distribution and highlighted potential issue. We will cover all of this plus our thoughts on how to solve the problems we encountered for each column in the next sections.

Prevalente

The "Prevalente" field was the field with the most null value reaching 92,5%. We inspected

further the "Prevalente" attribute and we discovered that there are many singleton and badly formatted values.

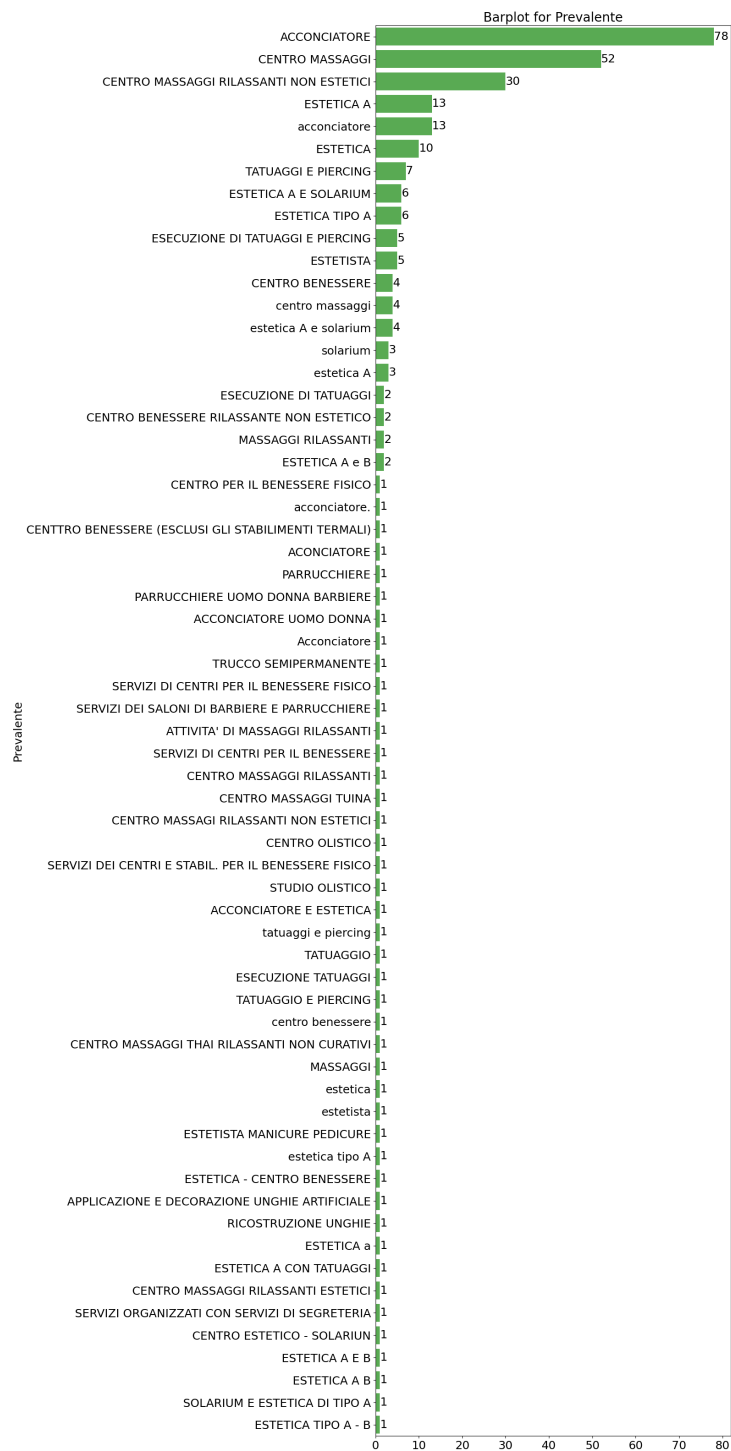


Figure 2.3: Enter Caption

Moreover for 52% of the value this is just a replica of the "Tipo esercizio pa" value. For these reasons we decided to merge it with the "Tipo esercizio pa" column and then drop

it. We will talk more about how we handled these columns in section 2.2.1.

Tipo esercizio PA:

In the "Tipo esercizio PA" column, multiple values are present, separated by a semicolon (;). For example, a shop that provides both hairdresser and beautician services will be marked as "ACCONCIATORE; ESTETISTA".

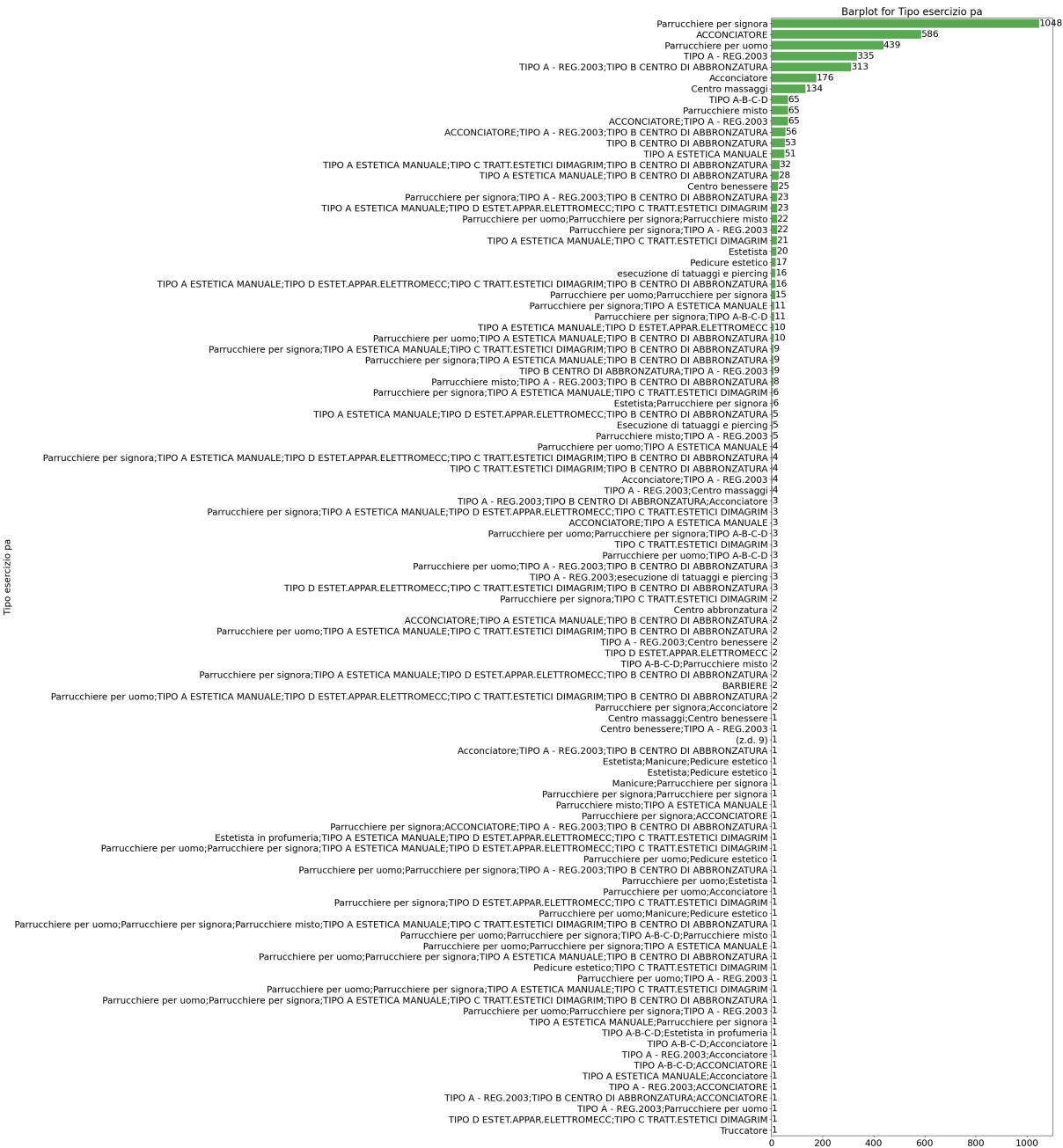


Figure 2.4: "Tipo esercizio PA" histogram

This format is suboptimal for readability and complicates statistical analysis of the column. Therefore, a key part of the data wrangling process will involve separating the

various service typologies and normalizing this attribute to reduce its entropy.

Tipo Via:

The high constancy was confirmed by the abundance of the value "VIA" compared to the other values:

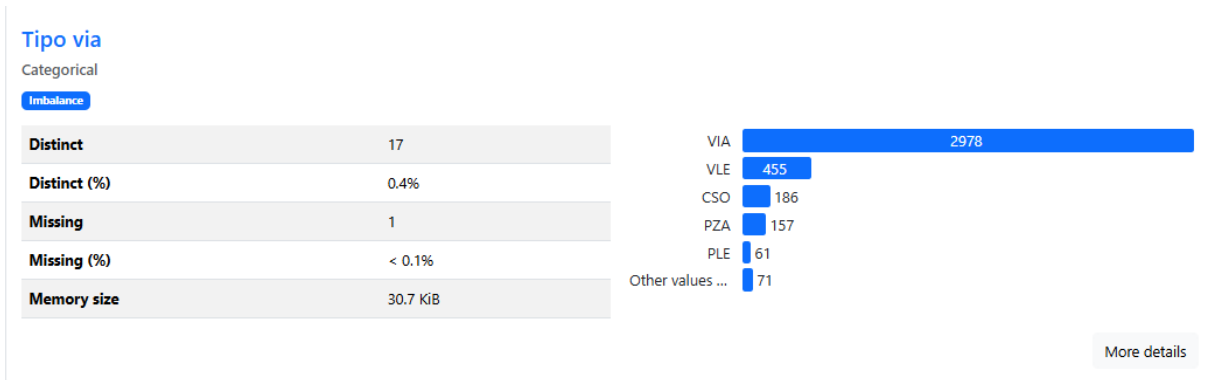


Figure 2.5: Tipo Via distribution

We verified the values that were not compliant with the official Milan toponymy dataset, available at the following ([link](#)), which contains all the official street name classifications. The following list represents the standardized three-letter codes for street types: ['PSG', 'ALZ', 'BST', 'CAV', 'CSO', 'FOR', 'GLL', 'GRD', 'LGO', 'MNT', 'PAR', 'PAS', 'PZA', 'PLE', 'PZT', 'PTE', 'PTA', 'RCA', 'RIP', 'SIT', 'STT', 'STV', 'SDA', 'SUP', 'TGZ', 'VIA', 'VPR', 'VLE', 'VLO', 'PSR', 'ROT', 'SPR', 'SNT'].

The analysis revealed that only 3 tuples (VIE and COMO) or 0.1% of the records contained invalid entries making the accuracy for this check of about 99,9%.

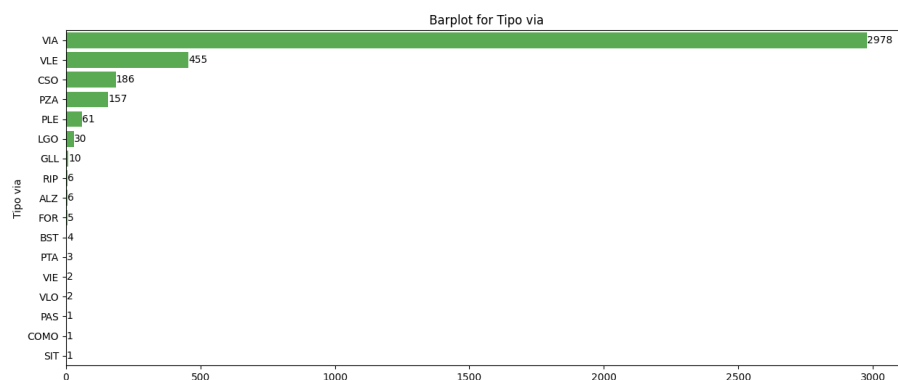


Figure 2.6: Tipo Via

Upon further investigation we found that the unknown values are:

- "VIE": probably "VIA" or "VLE".
- "COMO": probably the name of the "via" mistakenly put here.

In the data wrangling part we imputed them as "VIA" which is by far the most present value. We will deterministically impute these values also in a later step, exploiting the functional dependency "Via" $\rightarrow$ "Tipo via".

Via:

We checked how many activity there were in each via:

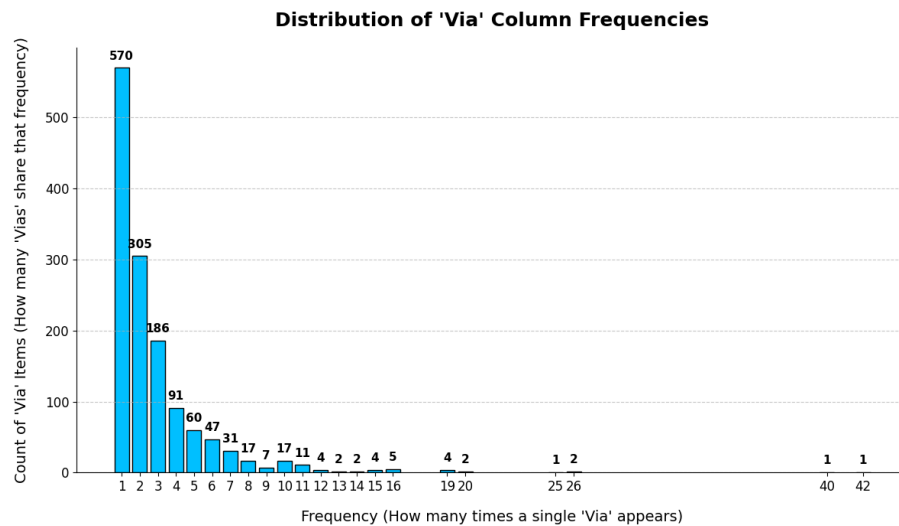


Figure 2.7: Number of activity for each Via

We found out that most of the vias we have have just 1 shop. We thought this might be due to some normalization issues. We imported a dataset scraped from the [link](#) containing all the "Via" of Milan searching for possible inconsistency. A part for a value of "15" inside this field, which we addressed in the data wrangling part, did not find anything concerning. As a double check, we also visually checked the "Via" that has the highest Levenshtein similarity (evaluated as $\text{Levenshtein Similarity} = 1 - (\text{Levenshtein Distance} / \text{Max Length})$). We found just one possible normalization issue (DI PTA TICINESE =? DI PORTA TICINESE) but after a more accurate analysis we found out that it is not since they have different "Codice Via" and one is a Corso and the other a "RIPA". We overall concluded that the Via field was basically not only complete for the most part but also normalized pretty well.

"Superficie lavorativa" and "Superficie altri usi"

Concerning the attributes "Superficie lavorativa" and "Superficie altri usi", we plotted the data to see if we can immediately identify some potential outliers.

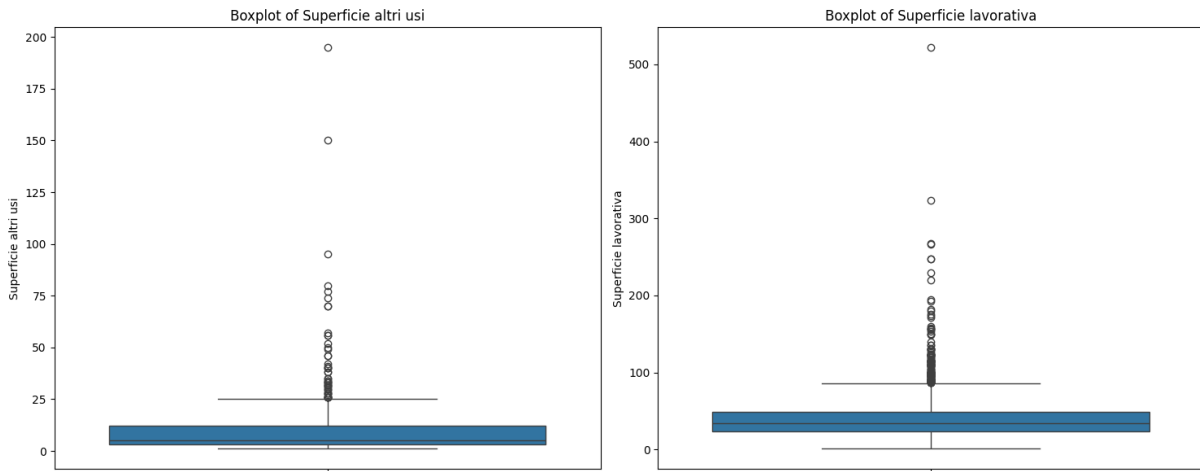


Figure 2.8: Boxplot "superfici"

From these boxplots, we can observe that some values deviate slightly from the general trend, but none appear to be extreme outliers. As shown here ([link](#)) one of the biggest hairdressing saloon in Milan is 300 meters squared so we will deal just with the value that have a total surface that is significantly higher than this value. We analyzed also both the value together.

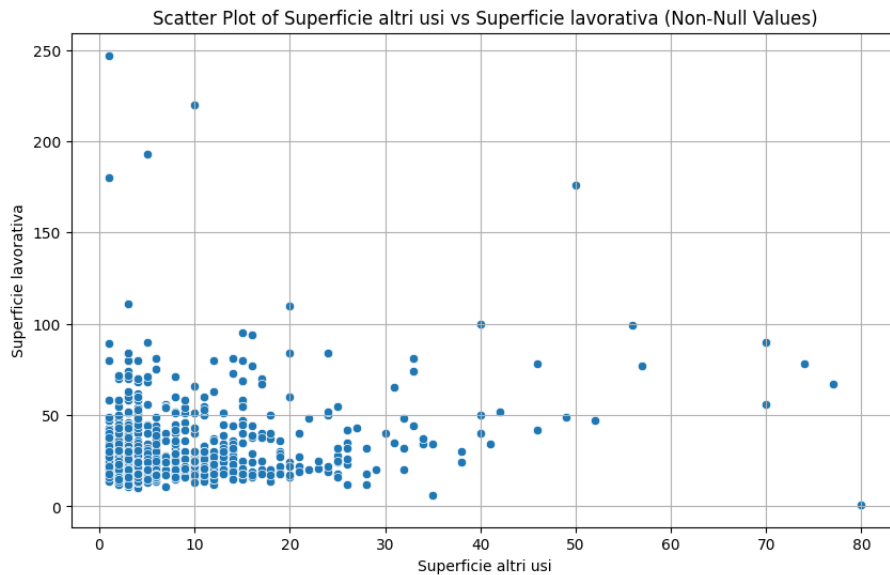


Figure 2.9: Plot of "superfici" columns

We hypothesized that the majority of the available space would generally be allocated primarily to the area open to the public rather than to areas reserved for employees. Our hypothesis was confirmed by the data. The results show that "Superficie lavorativa" (working surface) has an average value of 40.16, which is significantly higher than the 9.56 observed for "Superficie altri usi" (other uses surface). This is also shown clearly

in the graph above. In light of this, the value we found merging together the 2 columns that concerns us the most is the data point with 80 square meters for "Superficie altri usi" and 0 for "Superficie lavorativa." When we address outliers, we will certainly take a closer look at the specific store where this general trend of having a larger working area is not respected. We will analyze outliers so not only by per se but also considering the other surfaces if available (conditional outlier detection).

ZD:

The "ZD" column, which represents the Milan municipality code used to differentiate city areas (for example, zone 3 includes Città Studi, Lambrate, and Venezia), and accepts values from 1 to 9. The ZD seems to be decently evenly distributed.

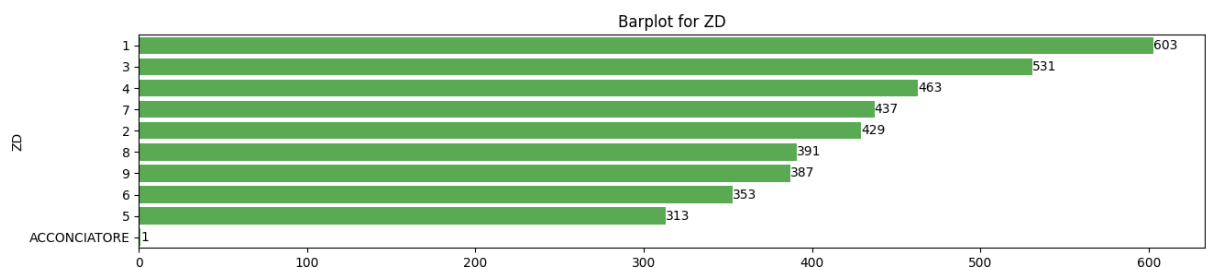


Figure 2.10: ZD histogram

The only value that is higher is the zd that refers to the center and this makes sense since it is the most populated area of Milan:

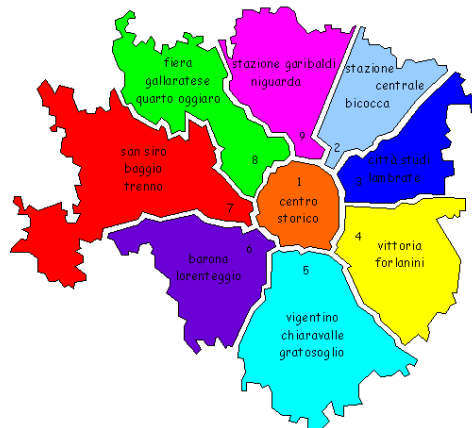


Figure 2.11: ZD milan

In 2.10 we can also see one wrong value in "ACCONCIATORE" which probably refers to another category and was there by accident. Since just 1 value is wrong we have an high syntactical accuracy of almost 99,97% for this column.

Civico:

After verifying that the column of Civico didn't contain any non-numeric values (eg: 12b), we cast it to int (anticipating a data wrangling operation) to show the value distribution:

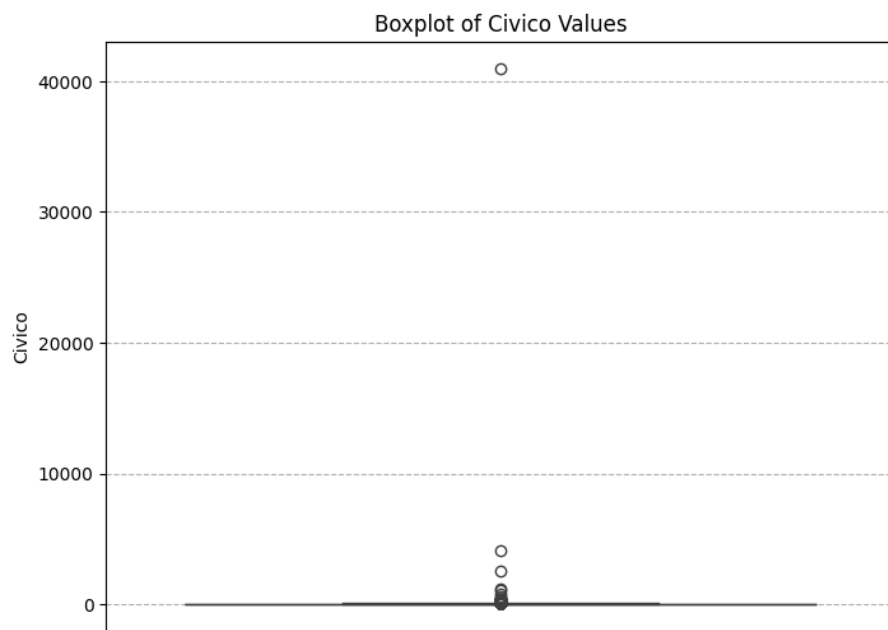


Figure 2.12: Civico boxplot

here we can see many outliers and values which doesn't make much sense which will be addressed in the outlier detection part.

Codice Via:

The "Codice via" attribute appears to be a numeric value ranging from 1 to 7548, which uniquely identifies each street specified in the "Via" column.

Codice via

Real number (ℝ)

High correlation

Distinct	1376	Minimum	1
Distinct (%)	35.2%	Maximum	7548
Missing	1	Zeros	0
Missing (%)	< 0.1%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	3752.8872	Memory size	30.7 KiB

Figure 2.13: Codice Via

We thought that it could thus follow Benford law but apparently it is not:

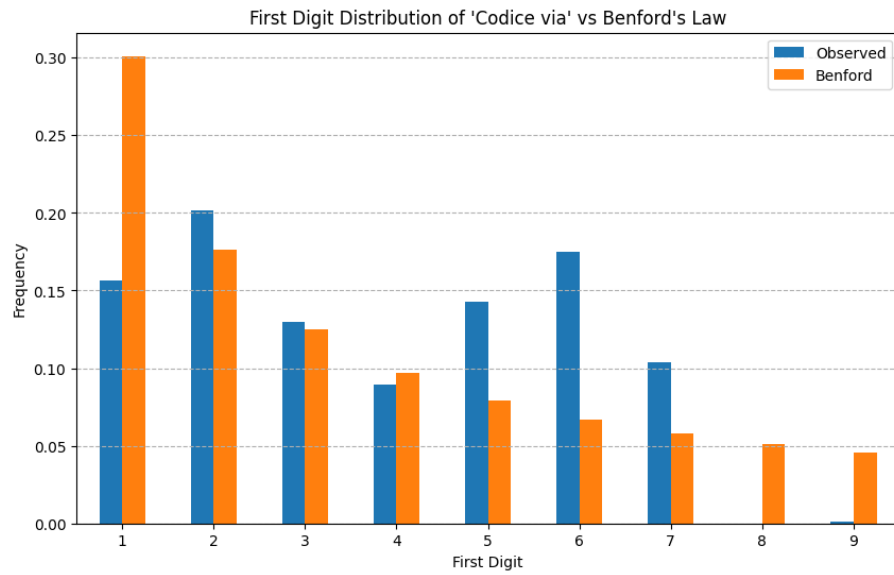


Figure 2.14: Benford law - Codice Via

Upon further investigation we realized that most of the values are in the first range of the 1000 spectrum (eg there are many values inside the 4000-4500 spectrum and basically none in the 4500-5000 one).

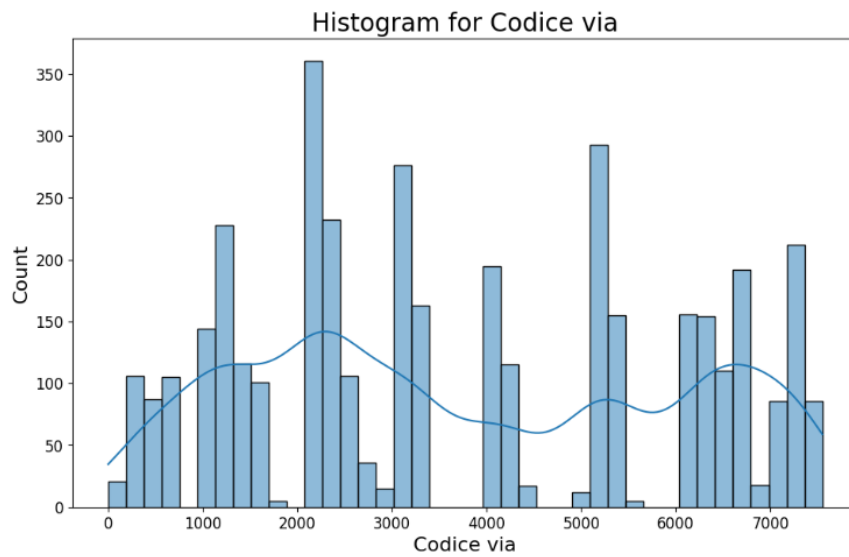


Figure 2.15: Codice Via distribution

we suspected this correlation might be due to a geographical numbering based on the ZD but we will confirm this later in the 2.1.2 section.

2.1.2. Correlation between columns

The report of y data profiling highlights a strong correlation between the fields "Codice via" and "ZD", as indicated by a high Pearson correlation coefficient. This relationship, clearly visible in the heat map, exceeds the defined threshold of 0.650. Such a correlation suggests a possible dependency between these two attributes.

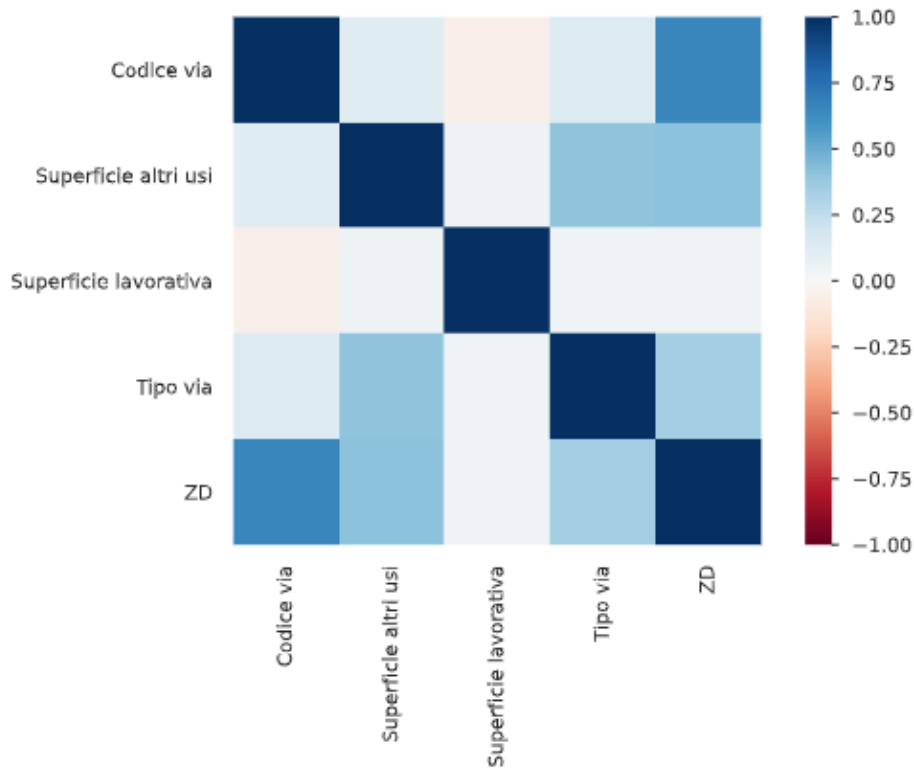


Figure 2.16: Correlation evaluation heatmap.

Upon further investigation we confirmed that the "Via" were assigned a number following a somewhat geographical order:

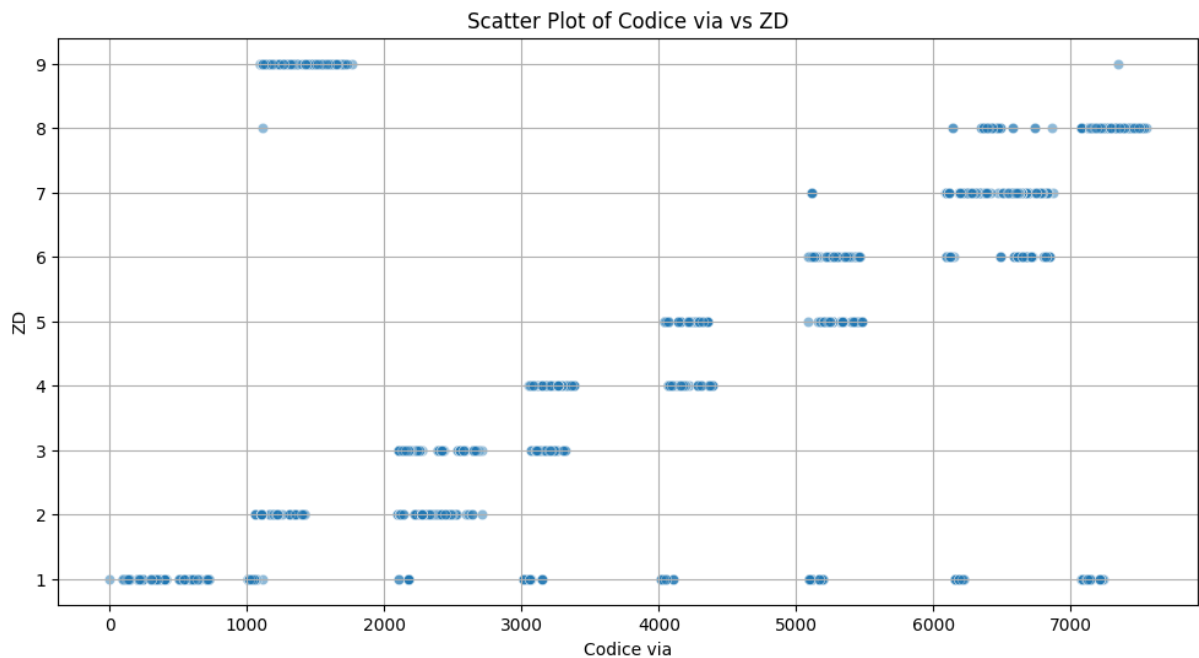


Figure 2.17: Codice Via VS ZD

From the plot 2.17 we can see that geographical near "ZD" also shares similar "Codice Via". The center, which borders with all the "ZD" has all sorts of "Codice Via", the 3rd ZD instead has part of its codice Via that overlap with the second ZD since are 2 regions which are near (and approximately in the 2000 values) and part in the 3000 values and most shared with the fourth ZD. Some outliers can be clearly observed in the data. Further analysis at the end of the cleaning stage will determine whether these values remain or are eliminated.

2.1.3. Perfect duplicate removal

Additionally, the profiling report indicates the presence of only one exact duplicate record in the dataset. This duplicate can be easily shown and then removed at the end of the data wrangling phase using the `drop()` function provided by the `pandas` library.

	Tipo esercizio pa	Ubicazione	Tipo via	Via	Civico	Codice via	ZD	Prevalente	Superficie altri usi	Superficie lavorativa
32	Acconciatore	CSO COMO N. 15 interno club f. conti	NaN	NaN	<NA>	NaN	NaN	NaN	NaN	NaN
33	(z.d. 9)	CSO	COMO	15	1111	9.0	ACCONCIATORE	NaN	195.0	NaN

Figure 2.18: Perfect duplicated row

2.1.4. Dependency discovery

Functional dependency

Via:

We have found the following functional dependencies:

- "Via" $\rightarrow$ "Codice via" for 99,56% of the cases
- "Codice Via" $\rightarrow$ "Via" for 100% of the cases
- "Via" $\rightarrow$ "Tipo Via" for 99,56% of the cases

Looking at the inconsistent tuples for which "Via" $\rightarrow$ "Tipo Via" does not hold we can say that there is a disproportionate values of one kind of "Tipo Via" rather than another, sign that the mistakes are random. In the next sections so we will impute the wrong values using the mode of the "Tipo via" for each via.

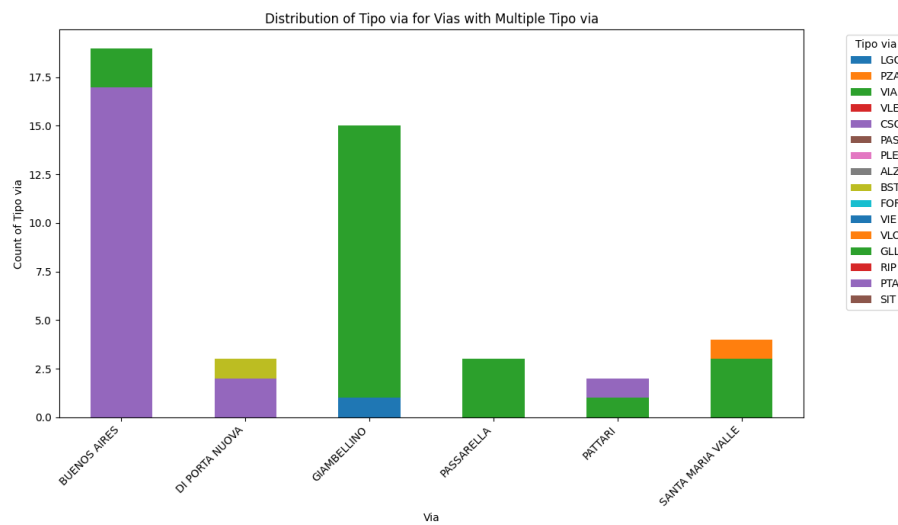


Figure 2.19: "Via" $\rightarrow$ "Tipo Via"

The same can be said about "Via" $\rightarrow$ "Codice via".

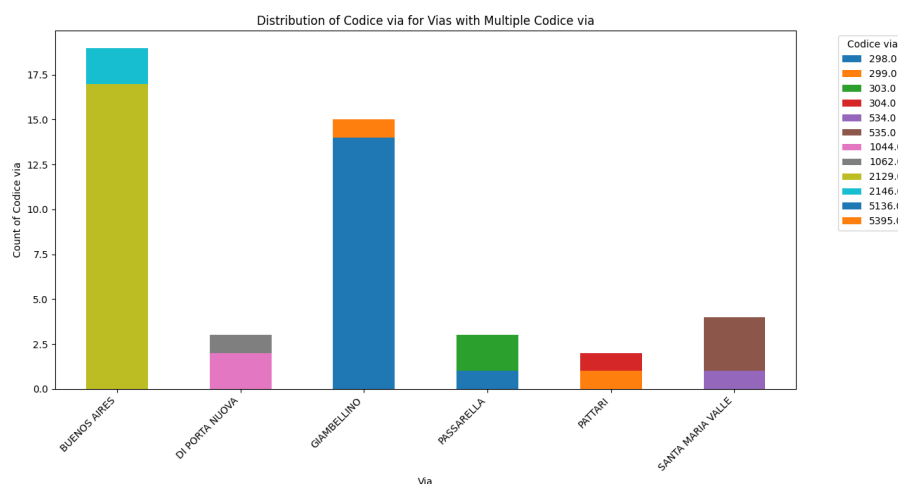


Figure 2.20: "Codice Via" $\rightarrow$ "Via"

Ubicazione:

We almost immediately recognized that the "Ubicazione" field also contained information

regarding other columns like "Tipo Via", "Via", "Civico" and "ZD". For instance, entries such as "VIA CORREGGIO N. 8 (z.d. 7)" contain these pieces of information combined in a single string. To keep the schema minimal, we needed to deal with this.

This functional dependency will be addressed in the data normalization part.

We decided to anticipate this activity, which should have been done in the normalization phase, because we needed to analyze the functional dependency between "Ubicazione" and these other columns and also get an idea of how much the final columns would differ. To address this, we applied a regular expression to extract the relevant components and created five new columns ("tipo", "via", "numero", "note", "zd"), allowing us to detect potential consistency issues within this attribute.

```
1 def estrai_dati_ubicazione(ubicazione):
2     if pd.isna(ubicazione):
3         return pd.Series([np.nan, np.nan, np.nan, np.nan, np.nan])
4     match = re.match(
5         r'^([A-Z]{2,})\s+',
6         r'(.+)\s+',
7         r'(?:(N\.|NUM\.)\s*',
8         r'([0-9]+)',
9         r'([A-Z0-9-./\s-]*)',
10        r'(?:(;|;|;)*\((Z\.\D\.\s*([0-9]+)\))?',
11        ubicazione.strip(),
12        flags=re.IGNORECASE
13    )
14    if match:
15        tipo_via, via, civico, interno, zd = match.groups()
16        civico = civico.lstrip('0') or '0'
17        interno = interno.strip() or np.nan
18        return pd.Series([
19            tipo_via.strip().upper(),
20            via.strip().upper(),
21            civico,
22            interno,
23            zd.strip() if zd else np.nan
24        ])
25    else:
26        return pd.Series([np.nan, np.nan, np.nan, np.nan, np.nan])
27
28 df[['Ubicazione tipo via', 'Ubicazione via', 'Ubicazione civico', 'Ubicazione interno',
    'Ubicazione zd']] = df['Ubicazione'].apply(estrai_dati_ubicazione)
```

Every attribute that we extracted from "Ubicazione" in the dataset is preceded by the word "Ubicazione" to distinguish it from the already present column, eg: "ZD" and "Ubicazione zd". Then we checked how many different tuple we had and we found out:

- "ZD" $\stackrel{99.56\%}{=}$ "Ubicazione zd"
- "Civico" $\stackrel{98.16\%}{=}$ "Ubicazione civico"
- "Tipo Via" $\stackrel{98.16\%}{=}$ "Ubicazione tipo via"
- "Via" $\stackrel{97.77\%}{=}$ "Ubicazione via"

In the next chapter, we will address these nearly duplicated columns by integrating them in the most effective way and handling potential conflicts appropriately.

2.1.5. Association rules

We also checked the dependency "Via" $\rightarrow$ "ZD". This FD have multiple tuple for which it doesn't hold and there are via for which we have an equal distribution of zd of one kind and zd of some other kind. An explanation for this is that there are Via which crosses 2 different z.d. An example of this is CSO LODI which crosses ZD 5 and ZD 6. While we can't say that this is so a Functional dependency this can be considered an association rule with a good support since this happens just for 45 "Via" out of the over 1300 distinct value of "via". So while it can't be used for performing error correction we can use this rule to reduce the number of null in the "ZD" columns using this association rule which seems to be accurate enough.

Key dependency

The columns "Via" and "Civico" should uniquely identify a store. In the initial dataset it is not like this unfortunately. The reasons why this could happen might be multiple:

- There was an expansion of the store.
- The store changed proprietary.
- The store was enlarged and took other responsibility (eg. passed to be just an hairdressing saloon to also offer other services like beautician services).

We will deal with this cases as potential duplicate taking into account these possible reasons.

2.1.6. Data quality assessment

We inspected 4 data quality measures, timeliness, completeness, consistency and accuracy.

Timeliness

We noted during the analysis that the dataset **lacks timestamp information**. Therefore, we will assume all data reflects the state at the time of the last update (**27/Nov/2012**).

Volatility is defined as a parameter relevant to the specific scope of the analysis. Given that we expect the number of activities to have **increased** in recent years and some activities to have **closed**, we anticipate that **volatility** will likely be **less than the currency** for most tasks.

The Timeliness score is calculated as:

$$\text{Timeliness} = \max \left\{ 0, 1 - \frac{\text{currency}}{\text{volatility}} \right\}$$

Consequently, as $\text{volatility} < \text{currency}$ will likely result in $\frac{\text{currency}}{\text{volatility}} > 1$, the formula returns

a timeliness score of zero (Timeliness = 0). Despite this, the data remains valuable for **historical analysis** and for mapping the general **distribution of hairdressing salons**.

Completeness

The dataset contains **3,909 rows**, which appears to be a reasonable quantity. This figure is supported by a 2014 article ([link](#)), which reported 6,762 open activities (hairdressers and beauticians) across the Province of Milan. Given this provincial total, the dataset size for the Municipality of Milan is plausible. Therefore, we will apply the **closed-world assumption** and treat the current data as a complete and accurate reflection of the target population for the purpose of assessing completeness.



Figure 2.21: CWA

From this assumption three attributes stand out as critical: "Prevalente", "Superficie altri usi", and "Superficie lavorativa", with missing values of 92.5%, 80.9%, and 33.5%, respectively. We will integrate the first column with "Tipologie PA" and try to impute the other 2 values to the best of our capabilities. We hope that by dealing with these problematic column we will higher the original **79%** of completeness of the whole dataset without compromising too much the accuracy.

Consistency

Regarding consistency we mainly checked the functional dependency that were found in the previous chapters of which mean is approximately **98,795%**.

Accuracy

Syntactical accuracy:

We checked the syntactical accuracy in the analysis we have done in the "ZD" field and in the "Tipo via" one. The mean for the 2 is approximately **99,35%**. **Semantical accuracy:**

2.2. Data Cleaning

2.2.1. Data normalization

The process of data normalization aims to define a set of common rules for the attributes in the dataset, in order to obtain a homogeneous and comparable set of values. In particular we performed the following activities:

1. **Manual intervention:** There were some rows with good data but in the completely wrong column. We decided to manually fix them immediately.
2. **Ubicazione:** Separated the "Ubicazione" field into different field to perform the functional dependency analysis as described in section 2.1.4
3. **Uppercase:** For textual fields, we decided to standardize all entries to uppercase to ensure consistency across the dataset. For example, in the column "Tipo esercizio pa", the values "ACCONCIATORE" and "Acconciatore" refer to the same category but are written differently.
4. **Casting Dtypes:** like Zd to integer.
5. **Removing invalid street types:** casted to null the invalid "Tipo via".
6. **Removing columns:** We removed the column titled "Prevalente" after copying its different values into the "Tipo esercizio pa" column, and removed the "Ubicazione" column after spreading those values across multiple columns. We also chose to remove the "Ubicazione Interno" column as it did not add relevant information.
7. **Normalizing "Tipo esercizio pa"** into four different column based on an importance order from "Primaria" to "Quaternaria".

Normalizing Tipo Pa

We began by splitting the semicolon-separated values into individual list elements, as previously described. Subsequently, we standardized the codes TIPO A, TIPO B, TIPO C, and TIPO D according to the following official mapping:

```
1 tipo_mapping = {  
2     'TIPO A': 'ESTETICA MANUALE',  
3     'TIPO B': 'CENTRO DI ABBRONZATURA',  
4     'TIPO C': 'TRATTAMENTI ESTETICI DIMAGRIMENTO',  
5     'TIPO D': 'ESTETISTA APPARECCHI ELETTROMECCANICI'  
6 }
```

Next, we imposed a cutoff on the number of services per shop, limiting them to a maximum of four. This decision resulted in modifying 32 shops that offered more than four services. This represents a negligible fraction of the **3,909** tuples in our dataset and it allowed us to allocate four dedicated columns for service-related data.

Through this restructuring, we transformed the dataset table into the First Normal Form (1NF). Specifically, the attribute `Tipo esercizio pa` was decomposed into four separate fields: `Tipologia primaria`, `Tipologia secondaria`, `Tipologia terziaria`, and `Tipologia quaternaria`.

It is worth noting that, although this design may increase data sparsity (as many entries in `Tipologia quaternaria` are null), we considered the trade-off advantageous.

The first normal form is essential to simplify the update and search queries and makes the whole schema more readable and usable. Moreover the null values do not represent missing data but rather accurately reflect the real-world situation where not all activities provide four distinct services.

Finally, we further reduced the entropy of this categorical attribute by merging synonymous labels (e.g., "CENTRO DI ABBRONZATURA" and "CENTRO ABBRONZATURA"). The complete list of substitutions applied is provided below:

```
1 substitution_map = {
2     'BARBIERE': 'PARRUCCHIERE PER UOMO',
3     'PARRUCCHIERE MISTO': 'ACCONCIATORE',
4     'PARRUCCHIERE PER SIGNORA': 'PARRUCCHIERE PER DONNA',
5     'MANICURE': 'ESTETISTA',
6     'ESTETISTA IN PROFUMERIA': 'ESTETISTA',
7     'CENTRO ABBRONZATURA': 'CENTRO DI ABBRONZATURA',
8     'PEDICURE ESTETICO': 'ESTETISTA',
9     'TRUCCATORE': 'ESTETISTA'
10 }
```

Reducing the entropy is a key step to simplify further possible data analysis tasks.

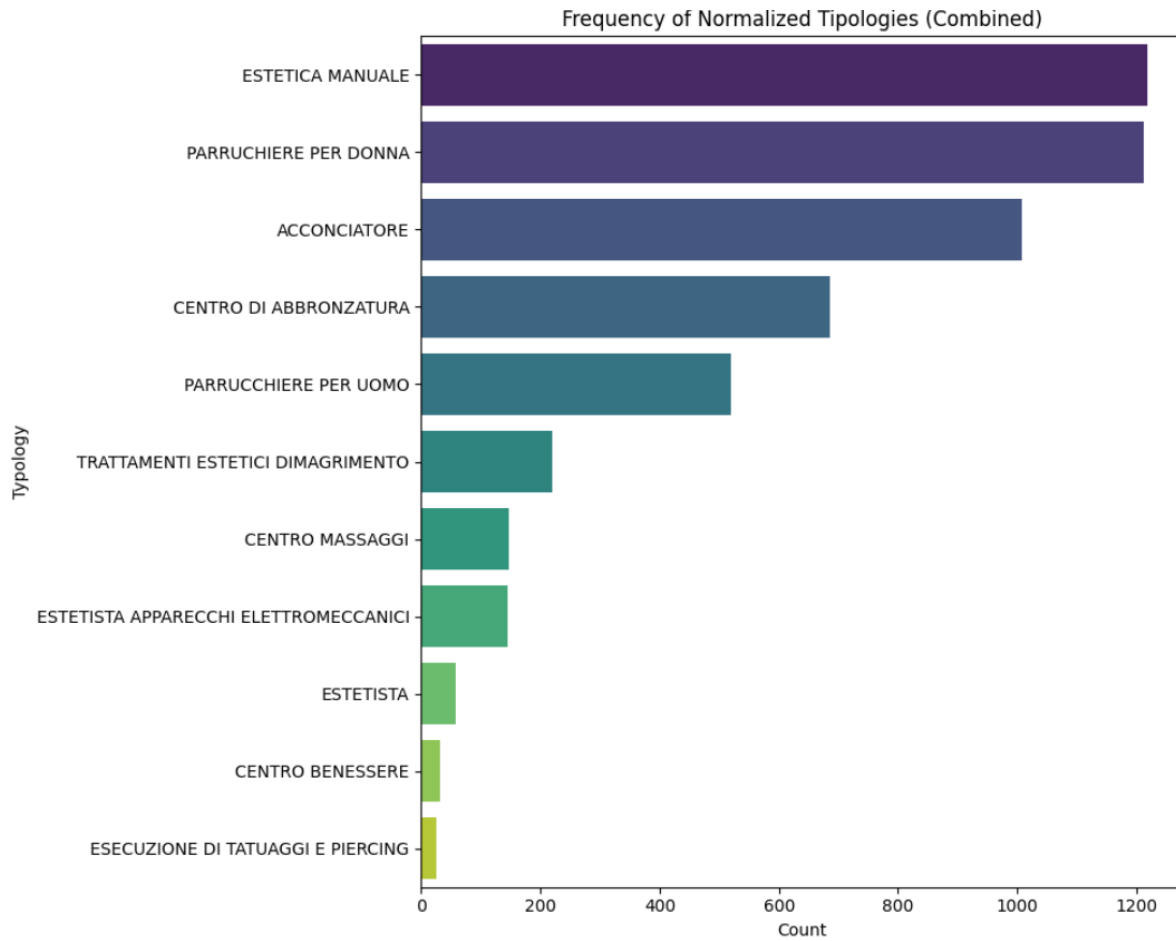


Figure 2.22: Tipologies

2.2.2. Error correction

Missing values handling

1. Deterministic techniques use predefined rules, such as functional dependencies (FD) or association rules.
2. Standard statistical imputation fills in missing values using simple aggregates, such as the average, median, mode, or minimum/maximum.
3. Machine Learning (ML) techniques use predictive models to estimate missing values based on patterns in the existing data.

We will use the deterministic approach as much as possible since it is the one that yields the best results and has the best explainability.

Deterministic imputation:

We used the duplicate columns that are coming from the splitting of "Ubicazione" in multiple columns to merge them together with the redundant columns we already have. We got marginal improvements on all columns except for "Tipo Via" where the improvement was significant.

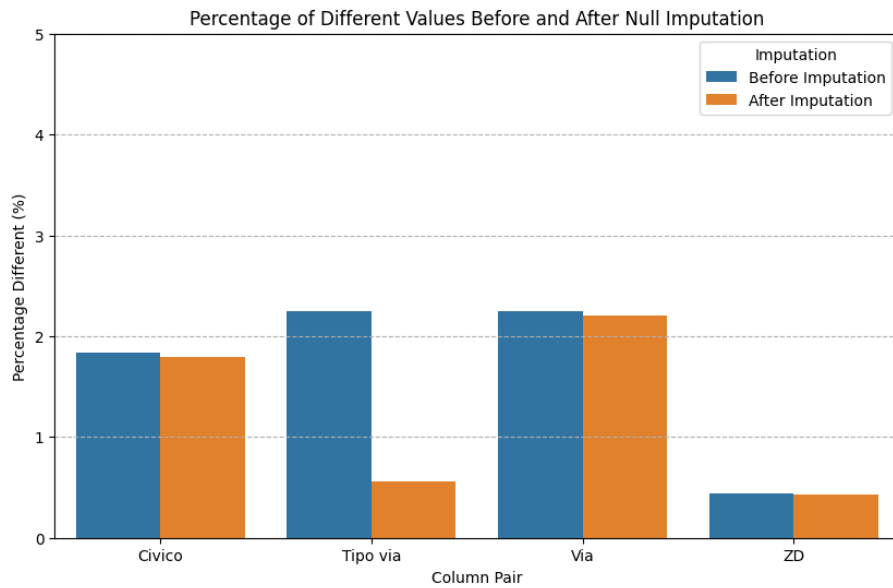


Figure 2.23: Deterministic imputation using "Ubicazione"

Standard statistical imputation

We used standard statistical imputation (mode) to address nulls that refers to the "Superficie" fields. Missing (NaN) values were filled using the mode of each category, grouped according to the "*Tipologia Primaria*" classification, as defined below:

```

1 categories_to_merge = {
2     "ACCONCIATORE": ["ACCONCIATORE", "PARRUCCHIERE PER UOMO", "PARRUCCHIERE PER DONNA"
3     ],
4     "ESTETICA": ["ESTETISTA", "ESTETICA MANUALE", "CENTRO DI ABBRONZATURA",
5     "TRATTAMENTI ESTETICI DIMAGRIMENTO", "ESTETISTA APPARECCHI
6     ELETTROMECCANICI"],
7     "BENESSERE": ["CENTRO MASSAGGI", "CENTRO BENESSERE"]
8 }

```

For activities with missing values in the "*Tipologia Primaria*" column, a new category named "**NOT DEFINED**" was introduced based on data quality theory, where missing categorical information can be explicitly labeled to preserve the integrity of the dataset and avoid introducing bias during analyses. The imputation relied on the mode of similar activities, based on the assumption that businesses within the same category tend to have comparable squared meters. **The completeness after the data imputation**

Column	Completeness (%)
Tipo via	100.00
Via	100.00
Civico	100.00
Codice via	100.00
ZD	100.00
Superficie altri usi	100.00
Superficie lavorativa	100.00
Tipologia Primaria	99.59
Tipologia Secondaria	22.93
Tipologia Terziaria	8.85
Tipologia Quaternaria	3.43

Table 2.2: Percentage of completeness in the column after the imputation

Functional Dependency Imputation

This section details the methods used to enforce functional dependencies (FDs) and resolve data inconsistencies, as previously outlined in Section 2.1.4.

Via:

To ensure data integrity, we enforced the FDs $\text{Via} \rightarrow \text{Tipo via}$ and $\text{Via} \rightarrow \text{Codice Via}$. In instances where conflicting values for **Tipo via** or **Codice Via** were found for the same **Via**, the conflict was resolved by imputing the statistical mode (most frequent value) associated with that specific **Via**.

Ubicazione:

Several imputation and cleaning steps were performed to resolve conflicts related to the **Ubicazione** field and its constituent attributes:

- **ZD $\rightarrow$ Ubicazione zd**: Inconsistencies for this dependency were addressed using the general association rule described in Section 2.1.5. For conflicting tuples, the correct ZD was determined by calculating the mode of all ZD values associated with that particular **Via**. It was observed that this computed mode always matched at least one of the two conflicting values (**ZD** or **Ubicazione zd**). We therefore applied this mode-based value, using a majority-vote approach, to impute the definitive ZD value.
- **Via $\rightarrow$ Ubicazione via**: Conflicts were first managed through normalization. Addresses with minor variations (e.g., MATTEOTTI GIACOMO vs. MATTEOTTI GIACOMO PIANO AMMEZZATO) were standardized, which reduced the conflicting tuples from 2.2% to 1.87%. A second, more complex category of conflicts involved genuinely different, yet valid, addresses (e.g., VIA FERMIGNANO 173 and VIA BOVISASCA 173). To resolve these, we leveraged the **Codice Via** attribute. We first verified that the **Codice Via** was significant (i.e., consistently associated with a single **Via** name). We then selected the **Via** name that corresponded to this verified **Codice Via**. After this automated process, only 2 tuples with typos in **Ubicazione via** remained,

which were corrected manually.

- **Tipo via $\rightarrow$ Ubicazione tipo via:** This functional dependency was automatically satisfied as a result of resolving the *Via $\rightarrow$ Ubicazione via* conflicts described in the previous step.
- **Civico $\rightarrow$ Ubicazione tipo civico:** A conflict rate of 1.76% persisted for this dependency, which could not be programmatically resolved. Given that the *Ubicazione* attributes consistently demonstrated the highest rate of inconsistencies and typographical errors across the dataset, we decided to prioritize the *Civico* attribute. Consequently, the *Ubicazione tipo civico* column was dropped.

2.2.3. Outliers detection

Detection on "Civico" column

To assess anomalies in the *Civico* column, we imported the dataset *civici_20251003.csv* from the official Milan toponymy portal [2]. A temporary dataframe containing only the relevant attribute was created in order to compute descriptive statistics (minimum, maximum, mean, standard deviation, most and least frequent values), which served as the reference domain for outlier identification.

- **Domain-based anomaly detection.** All values outside the observed range $\{\min, \max\}$ were flagged as anomalous street numbers.
- **Error correction strategy.** Detected anomalies were corrected by truncating the value to the first two digits (e.g., "40945" $\rightarrow$ "40"), assuming that errors typically occur in the less significant digits while leading digits remain reliable.

This procedure is grounded in data quality theory, which highlights several principles relevant to numeric attribute cleaning:

- **Error localization in numeric fields:** Typographical errors in numeric identifiers such as street numbers tend to occur in the less significant digits, while the leading digits are generally correct [3].
- **Consistency and domain conformity:** Truncating values to align with observed statistical ranges ensures conformity with the expected domain, a key criterion in standard data-quality frameworks (e.g., ISO 8000) [1].

After applying these operations, the "*Civico*" column was considered fully normalized and suitable for further analysis.

Detection on *Superficie altri usi* and *Superficie lavorativa*

The cleaning and normalization process was carried out through the following main steps:

- **Handling missing values.** Values from *Superficie altri usi* were transferred to *Superficie lavorativa* whenever the latter contained **NaN**. Upper bounds of 120 m^2 (working) and 35 m^2 (non-working) were then applied to remove implausible mea-

surements, reflecting average shop dimensions and supporting the principle of *completeness*.

- **Semantic consistency correction.** Whenever the non-working surface exceeded the working one, the two values were swapped. This ensures alignment with the realistic assumption that auxiliary areas are generally smaller than operational spaces.
- **Minimum regulatory compliance.** Values below legal or practical thresholds were raised to category-specific minima: 8 m^2 (working) and 3 m^2 (non-working) for hairdressing; 14 m^2 and 5 m^2 for beauty and massage services. This step enforces *accuracy*, reflecting minimum workspace standards (link).
- **Imputation of remaining NaN.** Any remaining missing values were filled using the mode of each category. For the NOT DEFINE group, the highest mode from the other categories was used to avoid introducing artificial bias.
- **Outlier detection and correction.** Outliers were identified through the 3σ rule and replaced with the corresponding category mode, preserving internal distribution coherence while removing extreme anomalies.

The cleaned dataset was then compared with the original distribution (Figure 2.9). Only points within the lower-left region of the plot were retained after the filtering process.

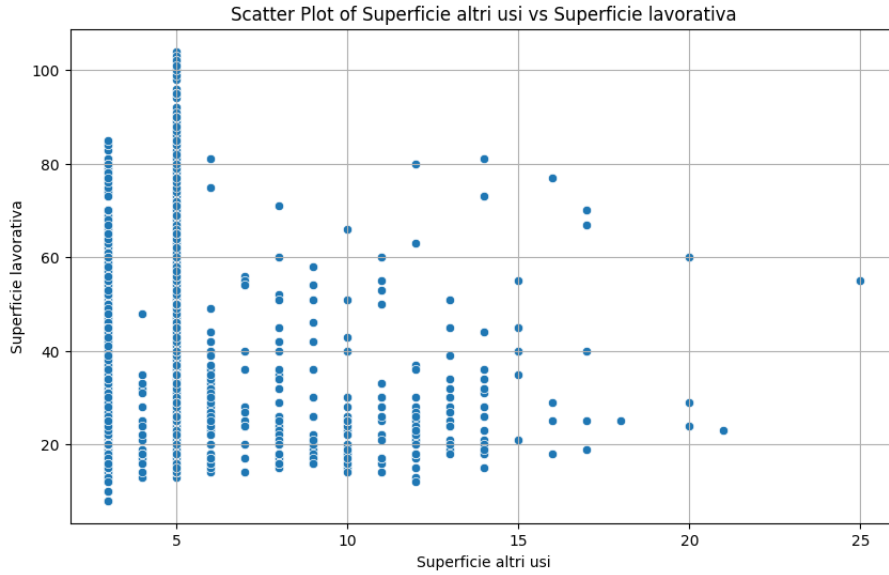


Figure 2.24: Plot of "superfici" columns

2.2.4. Duplicate detection

In the *duplication detection* phase, we aimed to identify records referring to the same physical activity or address. We employed a **knowledge-based technique**, leveraging the complete understanding of the dataset acquired during the functional dependency imputation described in the previous chapter. The combination $\{Tipo\ via, Via, Civico\}$ was selected as a super key based on the previous analysis which uniquely identifies each

location within the city of Milan and, consequently, each commercial activity. Records sharing the same key were grouped together and processed through a custom merging function to ensure consistent and reliable integration of their information. This approach significantly reduced data redundancy and improved the overall coherence of the dataset.

2.2.5. Data fusion

The assumption we made during data fusion was that stores enlarged either in terms of service provided or in terms of squared meters but a new tuple was inserted rather than updating the previous one. This seems the most probable explanation for our dataset. The following rules were applied to guarantee data integrity under this assumption during the merging:

- Typology consolidation:** Hierarchical consolidation of typologies was applied to maintain consistency and completeness of categorical information. By retaining the most specific typologies and removing redundant general ones, the dataset reflects the actual business characteristics without overrepresentation or ambiguity. This approach aligns with standard recommendations for categorical data management, ensuring that the dataset provides a clear and unambiguous representation of each commercial activity.
- Surface area integration:** For numeric attributes such as working and auxiliary surface areas, the choice to retain maximum values addresses potential underreporting and improves accuracy and reliability of quantitative data. This decision ensures that measurements used in subsequent analyses do not underestimate the true extent of the commercial space, which is critical for both spatial and business modeling.

Applying these principles systematically reduces redundancy and enhances coherence, an example is the following:

Tipo via	Via	Civico	Codice via	ZD	Sup. altri usi	Sup. lav.	Tip. Primaria	Tip. Secondaria	Tip. Terziaria	Tip. Quaternaria
VIA	DELLE FORZE ARMATE	234	6643.0	7	3.0	28.0	ACCONCIATORE	–	–	–
VIA	DELLE FORZE ARMATE	234	6643.0	7	5.0	50.0	ESTETISTA	–	–	–
VIA	DELLE FORZE ARMATE	234	6643.0	7	5.0	50.0	ESTETICA MANUALE	ESTETISTA APPARECCHI ELET-TROMECCANICI	TRATTAMENTI ESTETICI DIMAGRI-MENTO	CENTRO DI ABBRON-ZATURA
MERGED LINE										
VIA	DELLE FORZE ARMATE	234	6643.0	7	5.0	50.0	ACCONCIATORE	ESTETISTA	ESTETICA MANUALE	ESTETISTA APPARECCHI ELET-TROMECCANICI

Table 2.3: Original and merged rows for *Via Delle Forze Armate 234*

As shown in the dataframe, the three original records associated with the same geographical identifiers differ in their *Tipologia Primaria*. This variability indicates a temporal

evolution of the commercial activity, either through changes in ownership or in the services offered, resulting in multiple entries for the same physical location. Such patterns are typical in **longitudinal administrative datasets**, where new declarations accumulate without removing outdated ones.

A second observation concerns the relationship between *Superficie lavorativa* and the number of activity typologies. Larger working areas consistently correspond to a broader set of declared services, revealing a **positive correlation** between physical capacity and functional diversification. This supports the merging strategy by showing that records with **identical geographical identifiers** and **coherent spatial values** can be reliably consolidated into a single unit with consistent semantics.

After the merging procedure, the dataset was reduced to **3240 tuples**, implying that approximately **17%** of the initial entries were **duplicates** or **fragmented representations** of the same commercial unit. This reduction improves **structural consistency** and eliminates **redundancy** without loss of information, aligning with standard **Data Quality principles** [1].

The decrease in record count confirms that the harmonized dataset now represents only the activities **effectively active** at the time of the 2012 survey. Historical or outdated entries, often generated by **ownership transitions** or **service reclassification**, were consolidated into unified records. The resulting dataset provides a **cleaner, coherent, and analytically reliable** representation of the commercial landscape, suitable for subsequent modelling and spatial assessment.

df.head()

	Tipo esercizio pa	Ubicazione	Tipo via	Via	Civico	Codice via	ZD	Prevalente	Superficie altri usi	Superficie lavorativa
0	NaN	LGO DEI GELSOMINI N. 10 (z.d. 6)	LGO	DEI GELSOMINI	10	5394.0	6	NaN	NaN	55.0
1	NaN	PZA FIDIA N. 3 (z.d. 9)	PZA	FIDIA	3	1144.0	9	CENTRO MASSAGGI RILASSANTI NON ESTETICI	2.0	28.0
2	NaN	VIA ADIGE N. 10 (z.d. 5)	VIA	ADIGE	10	4216.0	5	CENTRO BENESSERE	2.0	27.0
3	NaN	VIA BARACCHINI FLAVIO N. 9 (z.d. 1)	VIA	BARACCHINI FLAVIO	9	356.0	1	TRUCCO SEMIPERMANENTE	NaN	NaN
4	NaN	VIA BERGAMO N. 12 (z.d. 4)	VIA	BERGAMO	12	3189.0	4	NaN	NaN	50.0

Figure 2.25: Dataframe head before the pipeline

df.head()

	Tipo via	Via	Civico	Codice via	ZD	Superficie altri usi	Superficie lavorativa	Tipologia Primaria	Tipologia Secondaria	Tipologia Terziaria	Tipologia Quaternaria
0	VIA	ABAMONTI GIUSEPPE	1	3135.0	3	6.0	16.0	PARRUCCHIERE PER UOMO	NaN	NaN	NaN
1	VIA	ABAMONTI GIUSEPPE	2	3135.0	3	5.0	20.0	PARRUCCHIERE PER UOMO	PARRUCCHIERE PER DONNA	ACCONCIATORE	ESTETICA MANUALE
2	VIA	ABBAZIA	6	1310.0	2	5.0	26.0	PARRUCCHIERE PER DONNA	NaN	NaN	NaN
3	VLE	ABRUZZI	16	2177.0	3	5.0	50.0	PARRUCCHIERE PER DONNA	ESTETICA MANUALE	CENTRO DI ABBRONZATURA	NaN
4	VLE	ABRUZZI	25	2177.0	3	12.0	22.0	PARRUCCHIERE PER DONNA	NaN	NaN	NaN

Figure 2.26: Dataframe head after the pipeline

3 | Conclusion and Final Assessment

The **data quality pipeline** applied throughout this project produced a substantial improvement in the **structural reliability**, **completeness**, and **internal coherence** of the dataset. All geographical and spatial attributes reached **100% completeness**, ensuring that every tuple now contains the full set of **location identifiers** and **size-related information**. This guarantees that the dataset is fully usable for **spatial**, **descriptive**, and **dependency-based analyses**.

The cleaning process also restored **consistency across functional dependencies**: as demonstrated in Chapter 2.1.4, the composite key *Tipo via + Via + Civico* now achieves a 100% uniqueness rate, confirming it as a **valid and stable identifier** for commercial units. This validates the correctness of **record merging** and reinforces the **structural soundness** of the final dataset.

The completeness level for *Tipologia Primaria* (**99.75%**) confirms that the primary classification is reliably available for nearly all shops. The lower completeness observed in secondary and higher-order typologies (*Secondaria*: **35.09%**, *Terziaria*: **14.91%**, *Quaternaria*: **5.86%**) is not due to missing data but reflects a deliberate **modelling choice**. These attributes were introduced to represent additional activity categories in a structured, hierarchical format, enabling **richer analytical interpretation** without compromising **data integrity**. Their values reflect the natural absence of additional typologies rather than data-quality issues.

From a **consistency** standpoint, the high **constancy rates** observed in typological attributes indicate the presence of recurring and semantically coherent classifications. This supports the reliability of the **typology restructuring**, demonstrating that the new hierarchy preserves meaningful patterns present in the raw data. Conversely, the low constancy in location attributes (e.g., *Via*: **1.14%**) is expected, reflecting the high territorial variability inherent in **address-based datasets**.

The resulting dataset is cleaner, fully consistent with its functional dependencies, and structurally aligned with standard **data quality dimensions**.

Overall, the final dataset provides a **robust** and **reliable foundation** for subsequent analytical tasks, ensuring that the information used for **modelling**, **categorization**, and **territorial assessment** faithfully reflects the underlying commercial landscape.

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